

Data Science & Artificial Intelligence



Miscellaneous

Discussion Notes

DDP – 01

By– Siddharth sir



#Q. Consider the state space as shown below:

S A B C D E G



Path: S->B->G

Cost: 8



Path: S->A->E->G

Cost: 15



Path: S->C->F->G

Cost: 7



Search is incomplete, path not found

Apply BFS algorithm, and find out the path and cost of path using BFS.

Assume that nodes are generated in alphabetical order.

#Q. Two random variables, X and Y, have a joint probability distribution given by $P(X = x, Y = y)$. The joint probabilities are as follows: $P(X = 1, Y = 1) = 0.2$, $P(X = 1, Y = 2) = 0.3$, $P(X = 2, Y = 1) = 0.4$ and $P(X = 2, Y = 2) = 0.1$. What is the probability that $Y = 1$ given $X = 1$?

A 0,1

B 0,2

C 0,4 ✓

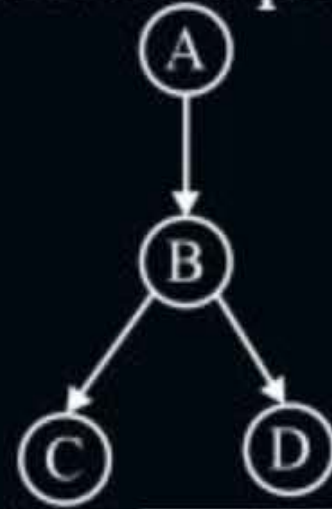
D 0,5

$$\begin{aligned} P(Y=1/X=1) &= \frac{P(Y=1, X=1)}{P(X=1)} \rightarrow \frac{0.2}{P(X=1, Y=1) + P(X=1, Y=2)} \\ &= \frac{0.2}{0.2 + 0.3} = \frac{2}{5} \Rightarrow 0.4 \end{aligned}$$

#Q. You have a bag of 20 marbles, 8 of which are red, and 12 are blue. You draw one marble from the bag, record its color (Event A), and then draw a second marble from the same bag without replacement and record its color (Event B). Calculate the probability that both marbles are red (Event A and Event B).

$$\begin{aligned} P(R_1, R_2) &= P(R_1) P(R_2|R_1) \\ &\quad \downarrow \\ &= \frac{8}{20} \cdot \frac{7}{19} = .147 \checkmark \end{aligned}$$

#Q. Given the following Bayesian Network consisting of four Bernoulli random variables and the associated conditional probability tables:



$$P(D=1/B=0) \quad P(C=0/B=0) \quad P(B=0/A=1) \quad P(A=1)$$

$$0.6 \times 1 \times 0.4 \times 0.4 \Rightarrow 0.096 \checkmark$$

	$P(\cdot)$
$A = 0$	0.6
$A = 1$	0.4

	$P(B = 0 \cdot)$	$P(B = 1 \cdot)$
$A = 0$	0.6	0.6
$A = 1$	0.4	0.4

	$P(C = 0 \cdot)$	$P(C = 1 \cdot)$
$B = 0$	1	0
$B = 1$	0	1

	$P(D = 0 \cdot)$	$P(D = 1 \cdot)$
$B = 0$	0.6	0.6
$B = 1$	1	1

The value of $P(A = 1, B = 0, C = 0, D = 1)$ _____. (Rounded off to three decimal places)

#Q. Consider the Bayesian Network shown below:



$$P(\text{Alarm1}) = 0.1$$

$$P(\text{Alarm2}) = 0.2$$

$$P(\text{Burglary} \mid \text{Alarm1}, \text{Alarm2}) = 0.8$$

$$P(\text{Burglary} \mid \text{Alarm1}, \neg \text{Alarm2}) = 0.7$$

$$P(\text{Burglary} \mid \neg \text{Alarm1}, \text{Alarm2}) = 0.6$$

$$P(\text{Burglary} \mid \neg \text{Alarm1}, \neg \text{Alarm2}) = 0.5$$

Calculate $P(\text{Alarm2} \mid \text{Burglary}, \text{Alarm1})$.

Enter ans upto two decimals

$$P(A_2 \mid B, A_1) = \frac{P(A_2, B, A_1)}{P(B, A_1)}$$

$$= P(B \mid A_2, A_1) P(A_2) P(A_1)$$

$$\Rightarrow 0.8 \times 0.2 \times 0.1 \Rightarrow 0.016$$

$$P(B, A_1) = \frac{P(B \mid A_1) P(A_1)}{P(B \mid A_1, A_2) P(A_2) + P(B \mid A_1, \neg A_2) P(\neg A_2)}$$

$$= \frac{P(B \mid A_1) P(A_1)}{P(B \mid A_1, A_2) P(A_2) + P(B \mid A_1, \neg A_2) P(\neg A_2)}$$

$$= 0.8 \times 0.2 + 0.7 \times 0.8$$

$$= 0.16 + 0.56$$

$$= 0.72$$

$$\text{So Ans} = \frac{0.16}{0.72} \Rightarrow 0.222$$

#Q. A Bayesian network consists of 4 binary variables. How many unique joint assignments can be generated using forward sampling from this network?

$$2^4 = 16$$

→ 3 variable = Total No of Case = 8, Truth = 5

#Q. Evaluate the number of truth assignments for the proposition $(P \vee Q) \rightarrow R$.

$(P \vee Q \rightarrow R)$ will be false if $P \vee Q = 1$ but $R = 0$

P	Q	R	
0	1	0	} Three Case this is zero
1	0	0	
1	1	0	

#Q. Simplify the following:

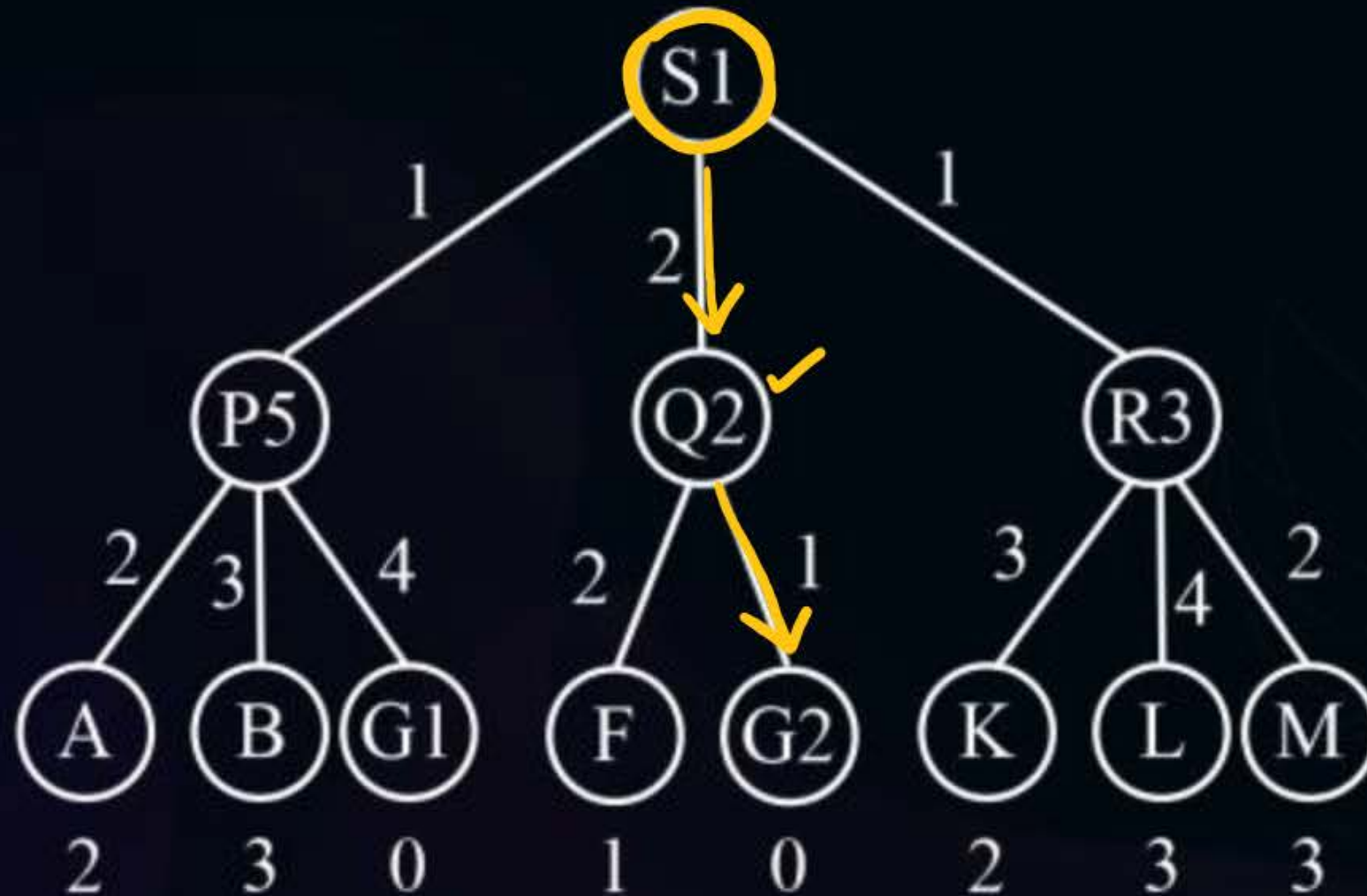
$$P \wedge (\underline{P \leftrightarrow Q}) \vee Q = Q$$

$$P \wedge (P \leftrightarrow Q) \vee Q$$

P	Q	$P \leftrightarrow Q$	$P \wedge (P \leftrightarrow Q)$	$\downarrow$ Q
0	0	1	0	0
0	1	0	0	1
1	0	0	0	0
1	1	1	1	1

#Q. The optional path cost from root to goal nodes G1 and G2 using GBFS is ____.
Use alphabetic order.

$\Downarrow$
heuristic



$(S \rightarrow Q \rightarrow G2)$

#Q. $\forall x [x \text{ node } 2 = 0, x \in \{2x^2 - 7x + 3 = 0\} \cap \{x^2 + 4x + 4 = 0\}] \Rightarrow \underline{x \text{ is empty}}$

$\nearrow 3, \frac{1}{2}$
 $\nearrow -2, -2$

~~S1:~~ The truth value of above expression is true.

S2: The truth value of above expression is False.

The number of correct statement is True.

$$x \in (2x^2 - 7x + 3 = 0)$$

$$2x^2 - 6x - x + 3 = 0$$

$$(x-3)(2x-1) = 0$$

$$3, \frac{1}{2}$$

$x = \text{domain} = \text{empty},$

$$x^2 + 4x + 4 = 0$$

$$(x+2)^2 =$$



THANK - YOU